

Deducing this

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§ 1 Abstract

We propose a new mechanism for specifying or deducing the value category of the expression that a member-function is invoked on. In other words, a way to tell from within a member function whether the expression it's invoked on is an lvalue or an rvalue; whether it is const or volatile; and the expression's type.

§ 2 Revision History

§ 2.1 Changes since r4

Wording and Implementation. Discussion about implicit vs explicit invocation and interaction with static functions.

§ 2.2 Changes since r3

The feedback from Belfast in EWG was “This looks good, come back with wording and implementation”. This version adds wording, the implementation is in the works.

§ 2.3 Changes since r2

[P0847R2] was presented in Kona in January 2019 to EWGI, with generally enthusiastic support.

This version adds:

- An FAQ entry for [library implementor feedback](#)
- An FAQ entry for [implementability](#).
- An FAQ entry for [computed deduction](#), an orthogonal feature that EWGI asked for in Kona.

§ 2.4 Changes since r1

[P0847R1] was presented in San Diego in November 2018 with a wide array of syntaxes and name lookup options.

Discussion there revealed some potential issues with regards to lambdas that needed to be ironed out. This revision zeroes in on one specific syntax and name lookup semantic which solves all the use-cases.

§ 2.5 Changes since r0

[P0847R0] was presented in Rapperswil in June 2018 using a syntax adjusted from the one used in that paper, using `this self&& self` to indicate the explicit object parameter rather than the `Self&& this self` that appeared in r0 of our paper.

EWG strongly encouraged us to look in two new directions:

- a different syntax, placing the object parameter's type after the member function's parameter declarations (where the *cv-ref* qualifiers are today)
- a different name lookup scheme, which could prevent implicit/unqualified access from within new-style member functions that have an explicit self-type annotation, regardless of syntax.

This revision carefully explores both of these directions, presents different syntaxes and lookup schemes, and discusses in depth multiple use cases and how each syntax can or cannot address them.

§ 3 Motivation

In C++03, member functions could have cv-qualifications, so it was possible to have scenarios where a particular class would want both a `const` and non-`const` overload of a particular member. (Note that it was also possible to want `volatile` overloads, but those are less common and thus are not examined here.) In these cases, both overloads do the same thing — the only difference is in the types being accessed and used. This was handled by either duplicating the function while adjusting types and qualifications as necessary, or having one overload delegate to the other. An example of the latter can be found in Scott Meyers’s “Effective C++” [EffCpp], Item 3:

```
class TextBlock {
public:
    char const& operator[](size_t position) const {
        // ...
        return text[position];
    }

    char& operator[](size_t position) {
        return const_cast<char&>(
            static_cast<TextBlock const&>(*this)[position]
        );
    }
    // ...
};
```

Arguably, neither duplication nor delegation via `const_cast` are great solutions, but they work.

In C++11, member functions acquired a new axis to specialize on: ref-qualifiers. Now, instead of potentially needing two overloads of a single member function, we might need four: `&`, `const&`, `&&`, or `const&&`. We have three approaches to deal with this:

- We implement the same member four times;
- We have three overloads delegate to the fourth; or
- We have all four overloads delegate to a helper in the form of a private static member function.

One example of the latter might be the overload set for `optional<T>::value()`, implemented as:

Quadruplication	Delegation to 4th	Delegation to helper
<pre>template <typename T> class optional { // ... constexpr T& value() & { if (has_value()) { return this->m_value; } throw bad_optional_access(); } constexpr T const& value() const& { if (has_value()) { return this->m_value; } throw bad_optional_access(); } constexpr T&& value() && { if (has_value()) { return move(this- >m_value); } } }</pre>	<pre>template <typename T> class optional { // ... constexpr T& value() & { return const_cast<T&>(static_cast<optional const&>(*this).value()); } constexpr T const& value() const& { if (has_value()) { return this->m_value; } throw bad_optional_access(); } constexpr T&& value() && { return const_cast<T&&>(static_cast<optional const&>(*this).value()); } }</pre>	<pre>template <typename T> class optional { // ... constexpr T& value() & { return value_impl(*this); } constexpr T const& value() const& { return value_impl(*this); } constexpr T&& value() && { return value_impl(move(*this)); } constexpr T const&& value() const&& { return value_impl(move(*this)); } private: template <typename Opt></pre>

<pre> throw bad_optional_access(); } constexpr T const&& value() const&& { if (has_value()) { return move(this->m_value); } throw bad_optional_access(); } // ... }; </pre>	<pre> constexpr T const&& value() const&& { return static_cast<T const&&>(value()); } // ... }; </pre>	<pre> static decltype(auto) value_impl(Opt&& opt) { if (!opt.has_value()) { throw bad_optional_access(); } return forward<Opt> (opt).m_value; } // ... }; </pre>
---	---	--

This is far from a complicated function, but essentially repeating the same code four times — or using artificial delegation to avoid doing so — begs a rewrite. Unfortunately, it’s impossible to improve; we *must* implement it this way. It seems we should be able to abstract away the qualifiers as we can for non-member functions, where we simply don’t have this problem:

```

template <typename T>
class optional {
    // ...
    template <typename Opt>
    friend decltype(auto) value(Opt&& o) {
        if (o.has_value()) {
            return forward<Opt>(o).m_value;
        }
        throw bad_optional_access();
    }
    // ...
};

```

All four cases are now handled with just one function... except it’s a non-member function, not a member function. Different semantics, different syntax, doesn’t help.

There are many cases where we need two or four overloads of the same member function for different `const`- or ref-qualifiers. More than that, there are likely additional cases where a class should have four overloads of a particular member function but, due to developer laziness, doesn’t. We think that there are enough such cases to merit a better solution than simply “write it, write it again, then write it two more times.”

§ 4 Proposal

We propose a new way of declaring non-static member functions that will allow for deducing the type and value category of the class instance parameter while still being invocable with regular member function syntax. This is a strict extension to the language.

We believe that the ability to write *cv-ref qualifier*-aware member function templates without duplication will improve code maintainability, decrease the likelihood of bugs, and make fast, correct code easier to write.

The proposal is sufficiently general and orthogonal to allow for several new exciting features and design patterns for C++:

- [recursive lambdas](#)
- a new approach to [mixins](#), a CRTP without the CRT
- [move-or-copy-into-parameter support for member functions](#)
- efficiency by avoiding double indirection with [invocation](#)

— perfect, sfinae-friendly [call wrappers](#)

These are explored in detail in the [examples](#) section.

This proposal assumes the existence of two library additions, though it does not propose them:

- `like_t`, a metafunction that applies the *cv*- and *ref*-qualifiers of the first type onto the second (e.g. `like_t<int&, double>` is `double&`, `like_t<X const&&, Y>` is `Y const&&`, etc.)
- `forward_like`, a version of `forward` that is intended to forward a variable not based on its own type but instead based on some other type. `forward_like<T>(u)` is short-hand for `forward<like_t<T, decltype(u)>>(u)`.

§ 4.1 Proposed Syntax

The proposed syntax in this paper is to use an explicit `this`-annotated parameter.

A non-static member function can be declared to take as its first parameter an *explicit object parameter*, denoted with the prefixed keyword `this`. Once we elevate the object parameter to a proper function parameter, it can be deduced following normal function template deduction rules:

```
struct X {
    void foo(this X const& self, int i);

    template <typename Self>
    void bar(this Self&& self);
};

struct D : X { };

void ex(X& x, D const& d) {
    x.foo(42);           // 'self' is bound to 'x', 'i' is 42
    x.bar();             // deduces Self as X&, calls X::bar<X&>
    move(x).bar();       // deduces Self as X, calls X::bar<X>

    d.foo(17);           // 'self' is bound to 'd'
    d.bar();             // deduces Self as D const&, calls X::bar<D const&>
}
```

Member functions with an explicit object parameter cannot be `static` or have *cv*- or *ref*-qualifiers.

A call to a member function will interpret the object argument as the first (`this`-annotated) parameter to it; the first argument in the parenthesized expression list is then interpreted as the second parameter, and so forth.

Following normal deduction rules, the template parameter corresponding to the explicit object parameter can deduce to a type derived from the class in which the member function is declared, as in the example above for `d.bar()`.

We can use this syntax to implement `optional::value()` and `optional::operator->()` in just two functions instead of the current six:

```
template <typename T>
struct optional {
    template <typename Self>
    constexpr auto&& value(this Self&& self) {
        if (!self.has_value()) {
            throw bad_optional_access();
        }

        return forward<Self>(self).m_value;
    }

    template <typename Self>
```

```
constexpr auto operator->(this Self&& self) {
    return addressof(self.m_value);
}
};
```

This syntax can be used in lambdas as well, with the `this`-annotated parameter exposing a way to refer to the lambda itself in its body:

```
vector captured = {1, 2, 3, 4};
[captured](this auto&& self) -> decltype(auto) {
    return forward_like<decltype(self)>(captured);
}

[captured]<class Self>(this Self&& self) -> decltype(auto) {
    return forward_like<Self>(captured);
}
```

The lambdas can either move or copy from the capture, depending on whether the lambda is an lvalue or an rvalue.

§ 4.2 Proposed semantics

What follows is a description of how deducing `this` affects all important language constructs — name lookup, type deduction, overload resolution, and so forth.

§ 4.2.1 Name lookup: candidate functions

In C++17, name lookup includes both static and non-static member functions found by regular class lookup when invoking a named function or an operator, including the call operator, on an object of class type. Non-static member functions are treated as if there were an implicit object parameter whose type is an lvalue or rvalue reference to `cv X` (where the reference and `cv` qualifiers are determined based on the function’s own qualifiers) which binds to the object on which the function was invoked.

For non-static member functions using an explicit object parameter, lookup will work the same way as other member functions in C++17, with one exception: rather than implicitly determining the type of the object parameter based on the `cv`- and `ref`-qualifiers of the member function, these are now explicitly determined by the provided type of the explicit object parameter. The following examples illustrate this concept.

C++17	Proposed
<pre>struct X { // implicit object has type X& void foo() &; // implicit object has type X const& void foo() const&; // implicit object has type X&& void bar() &&; };</pre>	<pre>struct X { // explicit object has type X& void foo(this X&); // explicit object has type X const& void foo(this X const&); // explicit object has type X&& void bar(this X&&); };</pre>

Name lookup on an expression like `obj.foo()` in C++17 would find both overloads of `foo` in the first column, with the non-const overload discarded should `obj` be `const`.

With the proposed syntax, `obj.foo()` would continue to find both overloads of `foo`, with identical behaviour to C++17.

The only change in how we look up candidate functions is in the case of an explicit object parameter, where the argument list is shifted by one. The first listed parameter is bound to the object argument, and the second listed parameter corresponds to the first argument of the call expression.

This paper does not propose any changes to overload *resolution* but merely suggests extending the candidate set to include non-static member functions and member function templates written in a new syntax. Therefore, given a call to `x.foo()`, overload resolution would still select the first `foo()` overload if `x` is not `const` and the second if it is.

The behaviors of the two columns are exactly equivalent as proposed.

The only change as far as candidates are concerned is that the proposal allows for deduction of the object parameter, which is new for the language.

Since in some cases there are multiple ways to declare the same function, it would be ill-formed to declare two functions with the same parameters and the same qualifiers for the object parameter. This is:

```
struct X {
    void bar() &&;
    void bar(this X&&); // error: same this parameter type

    static void f();
    void f(this X const&); // error: two functions taking no parameters
};
```

But as long as any of the qualifiers are different, it is fine:

```
struct Y {
    void bar() &;
    void bar() const&;
    void bar(this Y&&);
};
```

The rule in question is 12.2 [over.load]/2.2, and is extended in the wording below.

§ 4.2.2 Type deduction

One of the main motivations of this proposal is to deduce the cv-qualifiers and value category of the class object, which requires that the explicit member object or type be deducible from the object on which the member function is invoked.

If the type of the object parameter is a template parameter, all of the usual template deduction rules apply as expected:

```
struct X {
    template <typename Self>
    void foo(this Self&&, int);
};

struct D : X { };

void ex(X& x, D& d) {
    x.foo(1); // Self=X&
    move(x).foo(2); // Self=X
    d.foo(3); // Self=D&
}
```

It's important to stress that deduction is able to deduce a derived type, which is extremely powerful. In the last line, regardless of syntax, `Self` deduces as `D&`. This has implications for [name lookup within member functions](#), and leads to a potential [template argument deduction extension](#).

§ 4.2.3 By value this

But what if the explicit type does not have reference type? What should this mean?

```
struct less_than {
    template <typename T, typename U>
    bool operator()(this less_than, T const& lhs, U const& rhs) {
```

```

        return lhs < rhs;
    }
};

less_than{}(4, 5);

```

Clearly, the parameter specification should not lie, and the first parameter (`less_than{}`) is passed by value.

Following the proposed rules for candidate lookup, the call operator here would be a candidate, with the object parameter binding to the (empty) object and the other two parameters binding to the arguments. Having a value parameter is nothing new in the language at all — it has a clear and obvious meaning, but we’ve never been able to take an object parameter by value before. For cases in which this might be desirable, see [by-value member functions](#).

§ 4.2.4 Name lookup: within member functions

So far, we’ve only considered how member functions with explicit object parameters are found with name lookup and how they deduce that parameter. Now we move on to how the bodies of these functions actually behave.

Since the explicit object parameter is deduced from the object on which the function is called, this has the possible effect of deducing *derived* types. We must carefully consider how name lookup works in this context.

```

struct B {
    int i = 0;

    template <typename Self> auto&& f1(this Self&&) { return i; }
    template <typename Self> auto&& f2(this Self&&) { return this->i; }
    template <typename Self> auto&& f3(this Self&&) { return forward_like<Self>(*this).i; }
    template <typename Self> auto&& f4(this Self&&) { return forward<Self>(*this).i; }
    template <typename Self> auto&& f5(this Self&& self) { return forward<Self>(self).i; }
};

struct D : B {
    // shadows B::i
    double i = 3.14;
};

```

The question is, what do each of these five functions do? Should any of them be ill-formed? What is the safest option?

We believe that there are three approaches to choose from:

1. If there is an explicit object parameter, `this` is inaccessible, and each access must be through `self`. There is no implicit lookup of members through `this`. This makes `f1` through `f4` ill-formed and only `f5` well-formed. However, while `B().f5()` returns a reference to `B::i`, `D().f5()` returns a reference to `D::i`, since `self` is a reference to `D`.
2. If there is an explicit object parameter, `this` is accessible and points to the base subobject. There is no implicit lookup of members; all access must be through `this` or `self` explicitly. This makes `f1` ill-formed. `f2` would be well-formed and always return a reference to `B::i`. Most importantly, `this` would be *dependent* if the explicit object parameter was deduced. `this->i` is always going to be an `int` but it could be either an `int` or an `int const` depending on whether the `B` object is `const`. `f3` would always be well-formed and would be the correct way to return a forwarding reference to `B::i`. `f4` would be well-formed when invoked on `B` but ill-formed if invoked on `D` because of the requested implicit downcast. As before, `f5` would be well-formed.
3. `this` is always accessible and points to the base subobject; we allow implicit lookup as in C++17. This is mostly the same as the previous choice, except that now `f1` is well-formed and exactly equivalent to `f2`.

Following discussion in San Diego, the option we are proposing is #1. This allows for the clearest model of what a `this`-annotated function is: it is a `static` member function that offers a more convenient function call syntax. There is no implicit `this` in such functions, the only mention of `this` would be the annotation on the object parameter. All member access must be done directly through the object parameter.

The consequence of such a choice is that we will need to defend against the object parameter being deduced to a derived type. To ensure that `f5()` above is always returning a reference to `B::i`, we would need to write one of the following:

```
template <typename Self>
auto&& f5(this Self&& self) {
    // explicitly cast self to the appropriately qualified B
    // note that we have to cast self, not self.i
    return static_cast<like_t<Self, B>&&>(self).i;

    // use the explicit subobject syntax. Note that this is always
    // an lvalue reference - not a forwarding reference
    return self.B::i;

    // use the explicit subobject syntax to get a forwarding reference
    return forward<Self>(self).B::i;
}
```

§ 4.2.5 The Shadowing Mitigation / Private Inheritance Problem

The worst case for this proposal is the case where we do *not* intend on deducing a derived object - we only mean to deduce the qualifiers - but that derived type inherits from us privately and shadows one of our members:

```
class B {
    int i;
public:
    template <typename Self>
    auto&& get(this Self&& self) {
        // see above: we need to mitigate against shadowing
        return forward<Self>(self).B::i;
    }
};

class D : private B {
    double i;
public:
    using B::get;
};

D().get(); // error
```

In this example, `Self` deduces as `D` (not `B`), but our choice of shadowing mitigation will not work - we cannot actually access `B::i` from a `D` because that inheritance is private!

However, we don't have to rely on `D` to friend `B` to get this to work. There actually is a way to get this to work correctly and safely. C-style casts get a bad rap, but they are actually the solution here:

```
class B {
    int i;
public:
    template <typename Self>
    auto&& get(this Self&& self) {
        return ((like_t<Self, B>&&)self).i;
    }
};

class D : private B {
    double i;
public:
    using B::get;
};

D().get(); // error
```

No access checking for the win.

§ 4.2.6 Writing the function pointer types for such functions

As described in the previous section, the model for a member function with an explicit object parameter is a `static` member function.

In other words, given:

```
struct Y {
    int f(int, int) const&;
    int g(this Y const&, int, int);
};
```

While the type of `&Y::f` is `int(Y::*)(int, int) const&`, the type of `&Y::g` is `int(*) (Y const&, int, int)`. As these are *just* function pointers, the usage of these two member functions differs once we drop them to pointers:

```
Y y;
y.f(1, 2); // ok as usual
y.g(3, 4); // ok, this paper

auto pf = &Y::f;
pf(y, 1, 2); // error: pointers to member functions are not callable
(y.*pf)(1, 2); // okay, same as above
std::invoke(pf, y, 1, 2); // ok

auto pg = &Y::g;
pg(y, 3, 4); // okay, same as above
(y.*pg)(3, 4); // error: pg is not a pointer to member function
std::invoke(pg, y, 3, 4); // ok
```

The rules are the same when deduction kicks in:

```
struct B {
    template <typename Self>
    void foo(this Self&&);
};

struct D : B { };
```

Types are as follows:

- Type of `&B::foo<B>` is `void(*) (B&&)`
- Type of `&B::foo<B const&>` is `void(*) (B const&)`
- Type of `&D::foo<B>` is `void(*) (B&&)`
- Type of `&B::foo<D>` is `void(*) (D&&)`

This is exactly what happens if `foo` is a normal function.

By-value object parameters give you pointers to function in just the same way, the only difference being that the first parameter being a value parameter instead of a reference parameter:

```
template <typename T>
struct less_than {
    bool operator()(this less_than, T const&, T const&);
};
```

The type of `&less_than<int>::operator()` is `bool(*) (less_than<int>, int const&, int const&)` and follows the usual rules of invocation:

```
less_than<int> lt;
auto p = &less_than<int>::operator();

lt(1, 2);           // ok
p(lt, 1, 2);       // ok
(lt.*p)(1, 2);     // error: p is not a pointer to member function
invoke(p, lt, 1, 2); // ok
```

§ 4.2.7 Pathological cases

It is important to mention the pathological cases. First, what happens if `D` is incomplete but becomes valid later?

```
struct D;
struct B {
    void foo(this D const&);
};
struct D : B { };
```

Following the precedent of [P0929R2], we think this should be fine, albeit strange. If `D` is incomplete, we simply postpone checking until the point where we actually need a complete type, as usual. At that point `D().foo()` would be a valid expression. We see no reason to reject.

For unrelated complete classes or non-classes:

```
struct A { };
struct B {
    void foo(this A&);
    void bar(this int);
};
```

These are even more unlikely to be actually useful code. In this example, `B` is neither convertible to `A` nor `int`, so neither of these functions is even invocable using normal member syntax. However, they're still static member functions, so `B::bar(42)` is a valid call.

We think these declarations can best be left for compilers to warn about if they so choose, rather than coming up with a language rule to reject them.

Another interesting case, courtesy of Jens Maurer:

```
struct D;
struct B {
    int f1(this D);
};
struct D1 : B { };
struct D2 : B { };
struct D : D1, D2 { };

int x = D().f1(); // error: ambiguous lookup
int y = B().f1(); // error: B is not implicitly convertible to D
auto z = &B::f1; // ok
z(D());         // ok
B::f1(D{});     // ok
```

Even though both `D().f1()` and `B().f1()` are ill-formed, for entirely different reasons, taking a pointer to `&B::f1` is acceptable — its type is `int (*)(D)` — and that function pointer can be invoked with a `D`. Actually invoking this function does not require any further name lookup or conversion because by-value member functions do not have an implicit object parameter in this syntax (see [by-value this](#)). The same reasoning holds for the direct function invocation.

Again, we're not sure if these formulations are actually useful. More so that they don't seem harmful and attempting to reject these cases may accidentally reject useful ones.

§ 4.2.8 Teachability Implications

Explicitly naming the object as the `this`-designated first parameter fits within many programmers' mental models of the `this` pointer being the first parameter to member functions “under the hood” and is comparable to its usage in other languages, e.g. Python and Rust. It also works as a more obvious way to teach how `std::bind`, `std::thread`, `std::function`, and others work with a member function pointer by making the pointer explicit.

As such, we do not believe there to be any teachability problems.

§ 4.2.9 Can `static` member functions have an explicit object type?

No. Static member functions currently do not have an implicit object parameter, and therefore have no reason to provide an explicit one.

§ 4.2.10 Interplays with capturing `[this]` and `[*this]` in lambdas

Interoperability is perfect, since they do not impact the meaning of `this` in a function body. The introduced identifier `self` can then be used to refer to the lambda instance from the body.

§ 4.2.11 Parsing issues

The proposed syntax has no parsings issue that we are aware of.

§ 4.2.12 Code issues

There are two programmatic issues with this proposal that we are aware of:

1. Inadvertently referencing a shadowing member of a derived object in a base class `this`-annotated member function. There are some use cases where we would want to do this on purposes (see [crtp](#)), but for other use-cases the programmer will have to be aware of potential issues and defend against them in a somewhat verobse way.
2. Because there is no way to *just* deduce `const` vs non-`const`, the only way to deduce the value category would be to take a forwarding reference. This means that potentially we create four instantiations when only two would be minimally necessary to solve the problem. But deferring to a templated implementation is an acceptable option and has been improved by no longer requiring casts. We believe that the problem is minimal.

§ 5 Real-World Examples

What follows are several examples of the kinds of problems that can be solved using this proposal.

§ 5.1 Deduplicating Code

This proposal can de-duplicate and de-quadruplicate a large amount of code. In each case, the single function is only slightly more complex than the initial two or four, which makes for a huge win. What follows are a few examples of ways to reduce repeated code.

This particular implementation of `optional` is Simon's, and can be viewed on [GitHub](#). It includes some functions proposed in [\[P0798R0\]](#), with minor changes to better suit this format:

C++17	Proposed
<pre>class TextBlock { public: char const& operator[](size_t position) const { // ... }</pre>	<pre>class TextBlock { public: template <typename Self> auto& operator[](this Self&& self, size_t position) {</pre>

<pre> return text[position]; } char& operator[](size_t position) { return const_cast<char*>(static_cast<TextBlock const*> (this)[position]); } // ... }; </pre>	<pre> // ... return self.text[position]; } // ... }; </pre>
<pre> template <typename T> class optional { // ... constexpr T* operator->() { return addressof(this->m_value); } constexpr T const* operator->() const { return addressof(this->m_value); } // ... }; </pre>	<pre> template <typename T> class optional { // ... template <typename Self> constexpr auto operator->(this Self&& self) { return addressof(self.m_value); } // ... }; </pre>
<pre> template <typename T> class optional { // ... constexpr T& operator*() & { return this->m_value; } constexpr T const& operator*() const& { return this->m_value; } constexpr T&& operator*() && { return move(this->m_value); } constexpr T const&& operator*() const&& { return move(this->m_value); } constexpr T& value() & { if (has_value()) { return this->m_value; } throw bad_optional_access(); } constexpr T const& value() const& { if (has_value()) { return this->m_value; } throw bad_optional_access(); } constexpr T&& value() && { if (has_value()) { </pre>	<pre> template <typename T> class optional { // ... template <typename Self> constexpr like_t<Self, T&& operator*(this Self&& self) { return forward<Self>(self).m_value; } template <typename Self> constexpr like_t<Self, T&& value(this Self&& self) { if (this->has_value()) { return forward<Self>(self).m_value; } throw bad_optional_access(); } // ... }; </pre>

```

        return move(this->m_value);
    }
    throw bad_optional_access();
}

constexpr T const&& value() const&& {
    if (has_value()) {
        return move(this->m_value);
    }
    throw bad_optional_access();
}
// ...
};

```

```

template <typename T>
class optional {
    // ...
    template <typename F>
    constexpr auto and_then(F&& f) & {
        using result =
            invoke_result_t<F, T&>;
        static_assert(
            is_optional<result>::value,
            "F must return an optional");

        return has_value()
            ? invoke(forward<F>(f), **this)
            : nullopt;
    }

    template <typename F>
    constexpr auto and_then(F&& f) && {
        using result =
            invoke_result_t<F, T&&>;
        static_assert(
            is_optional<result>::value,
            "F must return an optional");

        return has_value()
            ? invoke(forward<F>(f),
                    move(**this))
            : nullopt;
    }

    template <typename F>
    constexpr auto and_then(F&& f) const& {
        using result =
            invoke_result_t<F, T const&>;
        static_assert(
            is_optional<result>::value,
            "F must return an optional");

        return has_value()
            ? invoke(forward<F>(f), **this)
            : nullopt;
    }

    template <typename F>
    constexpr auto and_then(F&& f) const&& {
        using result =
            invoke_result_t<F, T const&&>;
        static_assert(
            is_optional<result>::value,

```

```

template <typename T>
class optional {
    // ...
    template <typename Self, typename F>
    constexpr auto and_then(this Self&& self,
        F&& f) {
        using val = decltype((
            forward<Self>(self).m_value));
        using result = invoke_result_t<F, val>;

        static_assert(
            is_optional<result>::value,
            "F must return an optional");

        return this->has_value()
            ? invoke(forward<F>(f),
                    forward<Self>(self).m_value)
            : nullopt;
    }
    // ...
};

```

<pre> "F must return an optional"); return has_value() ? invoke(forward<F>(f), move(**this)) : nullopt; } // ... }; </pre>	
--	--

There are a few more functions in [\[P0798R0\]](#) responsible for this explosion of overloads, so the difference in both code and clarity is dramatic.

For those that dislike returning auto in these cases, it is easy to write a metafunction matching the appropriate qualifiers from a type. It is certainly a better option than blindly copying and pasting code, hoping that the minor changes were made correctly in each case.

§ 5.2 CRTP, without the C, R, or even T

Today, a common design pattern is the Curiously Recurring Template Pattern. This implies passing the derived type as a template parameter to a base class template as a way of achieving static polymorphism. If we wanted to simply outsource implementing postfix incrementation to a base, we could use CRTP for that. But with explicit objects that already deduce to the derived objects, we don't need any curious recurrence — we can use standard inheritance and let deduction do its thing. The base class doesn't even need to be a template:

C++17	Proposed
<pre> template <typename Derived> struct add_postfix_increment { Derived operator++(int) { auto& self = static_cast<Derived&> (*this); Derived tmp(self); ++self; return tmp; } }; struct some_type : add_postfix_increment<some_type> { some_type& operator++() { ... } }; </pre>	<pre> struct add_postfix_increment { template <typename Self> auto operator++(this Self&& self, int) { auto tmp = self; ++self; return tmp; } }; struct some_type : add_postfix_increment { some_type& operator++() { ... } }; </pre>

The proposed examples aren't much shorter, but they are certainly simpler by comparison.

§ 5.2.1 Builder pattern

Once we start to do any more with CRTP, complexity quickly increases, whereas with this proposal, it stays remarkably low.

Let's say we have a builder that does multiple things. We might start with:

```

struct Builder {
    Builder& a() { /* ... */; return *this; }
    Builder& b() { /* ... */; return *this; }
    Builder& c() { /* ... */; return *this; }
};

```

```
Builder().a().b().a().b().c();
```

But now we want to create a specialized builder with new operations `d()` and `e()`. This specialized builder needs new member functions, and we don't want to burden existing users with them. We also want `Special().a().d()` to work, so we need to use CRTP to *conditionally* return either a `Builder&` or a `Special&`:

C++17	Proposed
<pre>template <typename D=void> class Builder { using Derived = conditional_t<is_void_v<D>, Builder, D>; Derived& self() { return *static_cast<Derived*>(this); } public: Derived& a() { /* ... */; return self(); } Derived& b() { /* ... */; return self(); } Derived& c() { /* ... */; return self(); } }; struct Special : Builder<Special> { Special& d() { /* ... */; return *this; } Special& e() { /* ... */; return *this; } }; Builder().a().b().a().b().c(); Special().a().d().e().a();</pre>	<pre>struct Builder { template <typename Self> Self& a(this Self&& self) { /* ... */; return self; } template <typename Self> Self& b(this Self&& self) { /* ... */; return self; } template <typename Self> Self& c(this Self&& self) { /* ... */; return self; } }; struct Special : Builder { Special& d() { /* ... */; return *this; } Special& e() { /* ... */; return *this; } }; Builder().a().b().a().b().c(); Special().a().d().e().a();</pre>

The code on the right is dramatically easier to understand and therefore more accessible to more programmers than the code on the left.

But wait! There's more!

What if we added a *super*-specialized builder, a more special form of `Special`? Now we need `Special` to opt-in to CRTP so that it knows which type to pass to `Builder`, ensuring that everything in the hierarchy returns the correct type. It's about this point that most programmers would give up. But with this proposal, there's no problem!

C++17	Proposed
<pre>template <typename D=void> class Builder { protected: using Derived = conditional_t<is_void_v<D>, Builder, D>; Derived& self() { return *static_cast<Derived*>(this); } public: Derived& a() { /* ... */; return self(); } Derived& b() { /* ... */; return self(); } Derived& c() { /* ... */; return self(); } }; template <typename D=void> struct Special : Builder<conditional_t<is_void_v<D>, Special<D>, D> {</pre>	<pre>struct Builder { template <typename Self> Self& a(this Self&& self) { /* ... */; return self; } template <typename Self> Self& b(this Self&& self) { /* ... */; return self; } template <typename Self> Self& c(this Self&& self) { /* ... */; return self; } }; struct Special : Builder { template <typename Self> Self& d(this Self&& self) { /* ... */; return self; }</pre>


```

using Derived = typename
    Special::Builder::Derived;
Derived& d() { /* ... */; return this->self(); }
Derived& e() { /* ... */; return this->self(); }
};

struct Super : Special<Super>
{
    Super& f() { /* ... */; return *this; }
};

Builder().a().b().a().b().c();
Special().a().d().e().a();
Super().a().d().f().e();

```

```

template <typename Self>
Self& e(this Self&& self) { /* ...
    */; return self; }
};

struct Super : Special {
    template <typename Self>
    Self& f(this Self&& self) { /* ...
        */; return self; }
};

Builder().a().b().a().b().c();
Special().a().d().e().a();
Super().a().d().f().e();

```

The code on the right is much easier in all contexts. There are so many situations where this idiom, if available, would give programmers a better solution for problems that they cannot easily solve today.

Note that the Super implementations with this proposal opt-in to further derivation, since it's a no-brainer at this point.

§ 5.3 Recursive Lambdas

The explicit object parameter syntax offers an alternative solution to implementing a recursive lambda as compared to [P0839R0], since now we've opened up the possibility of allowing a lambda to reference itself. To do this, we need a way to *name* the lambda.

```

// as proposed in P0839
auto fib = [] self (int n) {
    if (n < 2) return n;
    return self(n-1) + self(n-2);
};

// this proposal
auto fib = [] (this auto self, int n) {
    if (n < 2) return n;
    return self(n-1) + self(n-2);
};

```

This works by following the established rules. The call operator of the closure object can also have an explicit object parameter, so in this example, `self` is the closure object.

In San Diego, issues of implementability were raised. The proposal ends up being implementable. See [the lambda FAQ entry](#) for details.

Combine this with the new style of mixins allowing us to automatically deduce the most derived object, and you get the following example — a simple recursive lambda that counts the number of leaves in a tree.

```

struct Leaf { };
struct Node;
using Tree = variant<Leaf, Node*>;
struct Node {
    Tree left;
    Tree right;
};

int num_leaves(Tree const& tree) {
    return visit(overload( // <-----+
        [](Leaf const&) { return 1; }, // |
        [](this auto const& self, Node* n) -> int { // |
            return visit(self, n->left) + visit(self, n->right); // <-----+
        })
    );
}

```

```
    ), tree);  
}
```

In the calls to `visit`, `self` isn't the lambda; `self` is the overload wrapper. This works straight out of the box.

§ 5.4 By-value member functions

This section presents some of the cases for by-value member functions.

§ 5.4.1 For move-into-parameter chaining

Say you wanted to provide a `.sorted()` method on a data structure. Such a method naturally wants to operate on a copy. Taking the parameter by value will cleanly and correctly move into the parameter if the original object is an rvalue without requiring templates.

```
struct my_vector : vector<int> {  
    auto sorted(this my_vector self) -> my_vector {  
        sort(self.begin(), self.end());  
        return self;  
    }  
};
```

§ 5.4.2 For performance

It's been established that if you want the best performance, you should pass small types by value to avoid an indirection penalty. One such small type is `std::string_view`. [Abseil Tip #1](#) for instance, states:

Unlike other string types, you should pass `string_view` by value just like you would an `int` or a `double` because `string_view` is a small value.

There is, however, one place today where you simply *cannot* pass types like `string_view` by value: to their own member functions. The implicit object parameter is always a reference, so any such member functions that do not get inlined incur a double indirection.

As an easy performance optimization, any member function of small types that does not perform any modifications can take the object parameter by value. Here is an example of some member functions of `basic_string_view` assuming that we are just using `charT` `const*` as iterator:

```
template <class charT, class traits = char_traits<charT>>  
class basic_string_view {  
private:  
    const_pointer data_;  
    size_type size_;  
public:  
    constexpr const_iterator begin(this basic_string_view self) {  
        return self.data_;  
    }  
  
    constexpr const_iterator end(this basic_string_view self) {  
        return self.data_ + self.size_;  
    }  
  
    constexpr size_t size(this basic_string_view self) {  
        return self.size_;  
    }  
  
    constexpr const_reference operator[](this basic_string_view self, size_type pos) {  
        return self.data_[pos];  
    }  
};
```

Most of the member functions can be rewritten this way for a free performance boost.

The same can be said for types that aren't only cheap to copy, but have no state at all. Compare these two implementations of `less_than`:

C++17	Proposed
<pre>struct less_than { template <typename T, typename U> bool operator()(T const& lhs, U const& rhs) { return lhs < rhs; } };</pre>	<pre>struct less_than { template <typename T, typename U> bool operator()(this less_than, T const& lhs, U const& rhs) { return lhs < rhs; } };</pre>

In C++17, invoking `less_than()(x, y)` still requires an implicit reference to the `less_than` object — completely unnecessary work when copying it is free. The compiler knows it doesn't have to do anything. We *want* to pass `less_than` by value here. Indeed, this specific situation is the main motivation for [\[P1169R0\]](#).

§ 5.5 SFINAE-friendly callables

A seemingly unrelated problem to the question of code quadruplication is that of writing numerous overloads for function wrappers, as demonstrated in [\[P0826R0\]](#). Consider what happens if we implement `std::not_fn()` as currently specified:

```
template <typename F>
class call_wrapper {
    F f;
public:
    // ...
    template <typename... Args>
    auto operator()(Args&&... ) &
        -> decltype(!declval<invoke_result_t<F&, Args...>>());

    template <typename... Args>
    auto operator()(Args&&... ) const&
        -> decltype(!declval<invoke_result_t<F const&, Args...>>());

    // ... same for && and const && ...
};

template <typename F>
auto not_fn(F&& f) {
    return call_wrapper<decay_t<F>>{forward<F>(f)};
}
```

As described in the paper, this implementation has two pathological cases: one in which the callable is SFINAE-unfriendly, causing the call to be ill-formed where it would otherwise work; and one in which overload is deleted, causing the call to fall back to a different overload when it should fail instead:

```
struct unfriendly {
    template <typename T>
    auto operator()(T v) {
        static_assert(is_same_v<T, int>);
        return v;
    }

    template <typename T>
    auto operator()(T v) const {
        static_assert(is_same_v<T, double>);
        return v;
    }
};
```

```

    }
};

struct fun {
    template <typename... Args>
    void operator()(Args&&...) = delete;

    template <typename... Args>
    bool operator()(Args&&...) const { return true; }
};

std::not_fn(unfriendly{})(1); // static assert!
                             // even though the non-const overload is viable and would be the
                             // best match, during overload resolution, both overloads of
                             // unfriendly have to be instantiated - and the second one is a
                             // hard compile error.

std::not_fn(fun{}());        // ok!? Returns false
                             // even though we want the non-const overload to be deleted, the
                             // const overload of the call_wrapper ends up being viable - and
                             // the only viable candidate.

```

Gracefully handling SFINAE-unfriendly callables is **not solvable** in C++ today. Preventing fallback can be solved by the addition of another four overloads, so that each of the four cv/ref-qualifiers leads to a pair of overloads: one enabled and one deleted.

This proposal solves both problems by allowing `this` to be deduced. The following is a complete implementation of `std::not_fn`. For simplicity, it makes use of `BOOST_HOF_RETURNS` from [Boost.HOF](#) to avoid duplicating expressions:

```

template <typename F>
struct call_wrapper {
    F f;

    template <typename Self, typename... Args>
    auto operator()(this Self&& self, Args&&... args)
        BOOST_HOF_RETURNS(
            !invoke(
                forward<Self>(self).f,
                forward<Args>(args)...))
};

template <typename F>
auto not_fn(F&& f) {
    return call_wrapper<decay_t<F>>{forward<F>(f)};
}

```

Which leads to:

```

not_fn(unfriendly{})(1); // ok
not_fn(fun{}());        // error

```

Here, there is only one overload with everything deduced together. The first example now works correctly. `Self` gets deduced as `call_wrapper<unfriendly>`, and the one `operator()` will only consider `unfriendly`'s non-`const` call operator. The `const` one is never even considered, so it does not have an opportunity to cause problems.

The second example now also fails correctly. Previously, we had four candidates. The two non-`const` options were removed from the overload set due to `fun`'s non-`const` call operator being `deleted`, and the two `const` ones which were viable. But now, we only have one candidate. `Self` is deduced as `call_wrapper<fun>`, which requires `fun`'s non-`const` call operator to be well-formed. Since it is not, the call results in an error. There is no opportunity for fallback since only one overload is ever considered.

This singular overload has precisely the desired behavior: working for unfriendly, and not working for fun.

This could also be implemented as a lambda completely within the body of `not_fn`:

```
template <typename F>
auto not_fn(F&& f) {
    return [f=forward<F>(f)](this auto&& self, auto&&... args)
        BOOST_HOF_RETURNS(
            !invoke(
                forward_like<decltype(self)>(f),
                forward<decltype(args)>(args)...))
        );
}
```

§ 6 Frequently Asked Questions

§ 6.1 On the implementability of recursive lambdas

In San Diego, 2018, there was a question of whether recursive lambdas are implementable. They are, details follow.

The specific issue is the way lambdas are parsed. When parsing a *non-generic* lambda function body with a default capture, the type of `this_lambda` would not be dependent, because the body is *not a template*. This leads to `sizeof(this_lambda)` not being dependent either, and must therefore have an answer - and yet, it cannot, as the lambda capture is not complete, and therefore, the type of `this_lambda` is not complete.

This is a huge issue for any proposal of recursive lambdas that includes non-generic lambdas.

Notice, however, that the syntax this paper proposes is the following:

```
auto fib = [](this auto&& self, int n) {
    if (n < 2) return n;
    return self(n-1) + self(n-2);
}
```

There is, quite obviously, no way to spell a non-generic lambda, because the lambda type is unutterable. `self`'s type is always dependent.

This makes expressions depending on `self` to be parsed using the regular rules of the language. Expressions involving `self` become dependent, and the existing language rules apply, which means both nothing new to implement, and nothing new to teach.

This proposal is therefore implementable, unlike any other we've seen to date. We would really like to thank Daveed Vandevoorde for thinking through this one with us in Aspen 2019.

§ 6.2 Would library implementers use this

In Kona, EWGI asked us to see whether library implementors would use this. The answer seems to be a resounding yes.

We have heard from Casey Carter and Jonathan Wakely that they are interested in this feature. Also, on the ewg/lewg mailing lists, this paper comes up as a solution to a surprising number of questions, and gets referenced in many papers-in-flight. A sampling of papers:

— [\[P0798R3\]](#)

— [\[P1221R1\]](#)

In Herb Sutter's "Name 5 most important papers for C++", 10 out of 289 respondents chose it. Given that the cutoff was 5, and that modules, throwing values, contracts, reflection, coroutines, linear algebra, and pattern matching were all in that

list, I find the result a strong indication that it is wanted.

We can also report that Gašper is dearly missing this feature in [libciabatta](#), a mixin support library, as well as his regular work writing libraries.

On the question of whether this would get used in the standard library interfaces, the answer was “not without the ability to constrain the deduced type”, which is a feature C++ needs even without this paper, and is an orthogonal feature. The same authors were generally very enthusiastic about using this feature in their implementations.

§ 6.3 Function Pointer Types

A valid question to ask is what should be the type of this-annotated functions that have a member function equivalent? There are only two options, each with a trade-off. Please assume the existence of these three functions:

```
struct Y {  
    int f(int, int) const&;           // exists  
    int g(this Y const&, int, int); // this paper  
    int h(this Y, int, int);         // this paper, by value  
};
```

`g` has a current equivalent (`f`), while `h` does not. `&Y::h`'s type *must* be a regular function pointer.

If we allow `g`'s type to be a pointer-to-member-function, we get non-uniformity between the types of `h` and `g`. We also get implementation issues because the types a template can result in are non-uniform (is this a template for a member function or a free function? Surprise, it's both!).

We also get forward compatibility with any conceivable proposal for extension methods - those will *also* have to be free functions by necessity, for roughly the same reasons.

The paper originally proposed it the other way, but this was changed to the current wording through EWG input in Cologne, 2018.

§ 6.4 Deducing to Base-Class Pointer

One of the pitfalls of having a deduced object parameter is when the intent is solely to deduce the cv-qualifiers and value category of the object parameter, but a derived type is deduced as well — any access through an object that might have a derived type could inadvertently refer to a shadowed member in the derived class. While this is desirable and very powerful in the case of mixins, it is not always desirable in other situations. Superfluous template instantiations are also unwelcome side effects.

One family of possible solutions could be summarized as **make it easy to get the base class pointer**. However, all of these solutions still require extra instantiations. For `optional::value()`, we really only want four instantiations: `&`, `const&`, `&&`, and `const&&`. If something inherits from `optional`, we don't want additional instantiations of those functions for the derived types, which won't do anything new, anyway. This is code bloat.

This is already a problem for free-function templates: The authors have heard many a complaint about it from library vendors, even before this paper was introduced, as it is desirable to only deduce the ref-qualifier in many contexts. Therefore, it might make sense to tackle this issue in a more general way. A complementary feature could be proposed to constrain *type deduction*.

The authors strongly believe this feature is orthogonal. However, hoping that mentioning that solutions are in the pipeline helps gain consensus for this paper, we mention one solution here. The proposal is in early stages, and is not in the pre-belfast mailing. It will be present in the post-belfast mailing: [computed deduction](#)

§ 7 Implementation

This has been implemented in the EDG front end, with gracious help and encouragement from Daveed Vandevoorde. Implementation didn't turn up any notable issues.

§ 8 Not quite static, not quite non-static

Where previously, member functions were divided into static member functions and non-static member functions, this gets a little more complex because some static member functions still use the implied object parameter (those that have an explicit `this` parameter) and some do not. This wording introduces the term “object member function” for the union of non-static member functions and static member functions with an explicit `this` parameter. Many functions were previously restricted to be non-static member functions are now restricted to be object member functions.

The biggest issue in working through the wording for this paper is: just *how* static is a function with an explicit `this` parameter? Member functions with an explicit `this` parameter are sort of half-static, half-non-static. They're static in the sense that a pointer to a function with an explicit `this` parameter has pointer-to-function type, not pointer-to-member type, and there is no implicit `this` in the body of such functions. They're non-static in the sense that class access to such a function must be a full call expression, and you can use such functions to declare operators:

```
struct C {
    void nonstatic_fun();

    int explicit_fun(this C c) {
        nonstatic_fun(); // error
        c.nonstatic_fun(); // ok
        static_fun(); // ok
        auto x = this; // error
        return 0;
    }

    static void static_fun() {
        explicit_fun(); // error
        explicit_fun(C{}); // error
        C{}.explicit_fun(); // ok
    }

    static void operator()(int); // error
    void operator()(this C, char); // ok
};

C c;
int (*a)(C) = &C::explicit_fun; // ok

auto x = c.static_fun; // ok
auto y = c.explicit_fun; // error
auto z = c.explicit_fun(); // ok
```

§ 8.1 Implicit this access

One question came up over the course of implementing this feature which was whether or not there should be implicit `this` syntax for invoking an “explicit `this`” member function from an “implicit `this`” one. That is:

```
struct D {
    void explicit_fun(this D const&);

    void implicit_fun() const {
        this->explicit_fun(); // obviously ok
        explicit_fun(); // but what about this?
    }
};
```

It's tempting to say that given an *explicit* `this` parameter, we should always require an *explicit* `this` argument. But one of the major advantages of having the “implicit this” call support in this context is that it allows derived types to not have to know about the implementation choice. As frequently used as a motivating example, we would like `std::optional` to be able to implement its member functions using this language feature if it wants to, without us having to know about it:

```
struct F : std::optional<int>
{
    bool is_big() const {
        // if this language feature required explicit call syntax, then this
        // call would be valid if and only if the particular implementation of
        // optional did not use this feature. This is undesirable
        return value() > 42;
    }
};
```

Or, more generally, the implementation strategy of a particular type's member function should not be relevant for users deriving from that type. So “implicit this” stays.

§ 8.2 Implied object argument

The way we talk about what it means to invoke a member function, there's always an object argument — whether explicit or implied. As mentioned above, it's important that this works:

```
struct A {
    void f(this A const&);
    void g() {
        f(); // #1
    }
};
```

Where the call in `#1` is equivalent to `(*this).f()`; due to the implicit `this`, that's the implied object argument.

But we say that `f` is a `static` member function, so what about this:

```
struct B {
    void f(this B const&);
    static void h() {
        f(B{}); // #2
    }
};
```

There is no `this` in a `static` member function, but there is *always* an object argument anyway - in this case, in general, we create a “contrived” argument of type `B` and then say the call fails if we end up picking a non-`static` member function that actually needs a real object argument. But that's the status quo, what should we do in this case? There are two options to consider:

1. We still create a contrived object argument (as we would if `f` were a “normal” `static` member function). But in this case, because `f` has an explicit `this` parameter, our contrived object argument would bind to the `this` parameter and this call would fail. That is, we're passing two arguments (a contrived object of type `B` and a `B`) to a function that takes a single parameter (`B const&`).
2. We can choose to *not* create a contrived object argument here, knowing that `f` has an explicit `this` parameter. As such, we're just invoking a single-parameter function with a single argument and this works fine.

It's tempting to say, sure, let's just choose option 2, since we would be allowing more functionality. But that runs into a wrinkle:

```
struct C {
    static void f(C&); // #3
    void f(this C const&); // #4
```



```
static void j() {
    C c;
    f(c); // #5
}
};
```

What does the call to `f(c)` do? #3 is our legacy C++20 (or even C++98) candidate. But the way this candidate works is that `f` behaves like a two-parameter function (the first parameter is ignored and matches everything equivalently and the second is `C&`) and we are providing two arguments to it (a contrived object of type `C` and an lvalue of type `C`). If we say that #2 above is a valid call, then #4 here would likewise also be a valid candidate - but here we only have a single-parameter function with a single argument (there is no contrived object argument or fake parameter). In other words, we're performing overload resolution now with candidates of different arity? This would be a first in C++.

It's probably doable to come up with wording to properly handle this scenario, ending up with #3 being invoked, but it seems fairly complicated and we're not even sure that we want to support the `f(B{})` or `f(c)` calls directly to begin with.

Instead, we propose *not* to support those calls. That is, while member functions with an explicit `this` parameter are considered static member functions, they must still be invoked with an object argument (which can be implied) just like non-static member functions.

This does come with its own quirk. Since a function with an explicit `this` parameter is still a `static` member function, a pointer to it still has function pointer type, which means you can invoke through the function pointer. Just not directly. That is:

```
struct B {
    void f(this B const&);
    static void h() {
        B{}.f(); // okay
        f(B{}); // proposed: error. The B{} has to be the object
                // argument, not just any kind of argument
    }
};

void external() {
    auto p = &B::f; // p is a void(*) (B const&)
    p(B{});         // okay
}
```

This is, admittedly, a weird place to arrive at.

§ 9 Proposed Wording

§ 9.1 Overview

The status quo here is that a member function has an *implicit object parameter*, always of reference type, which the *implied object argument* is bound to. The obvious name for what this paper is proposing is, then, the *explicit object parameter*. The problem with these names is: well, what is an object parameter? A parameter that takes an object? Isn't that most parameters?

Instead, the wording introduces the term *this parameter*, renaming implicit object parameter to implicit this parameter, and introducing the notion of an explicit this parameter. Alternate terms considered were "selector parameter" or "instance parameter".

§ 9.2 Wording

[Editor's note: This paper introduces many new terms that are defined in [dcl.dcl] - so even though the wording here is presented in standard layout order (we obviously want to ensure that `is_standard_layout<P0847>` is true), it may be helpful to refer to [those definitions](#) when reviewing the wording]

§ 9.2.1 Wording in 7 [expr]

Move 11.4.3 [class.mfct.non-static]/3 in front of 7.5.4 [expr.prim.id]/2 (the highlighted diff is relative to the original paragraph):

- ^{2*} When an *id-expression* that is not part of a class member access syntax and not used to form a pointer to member (7.6.2.2 [expr.unary.op]) is used in a member of class *X* in a context where `this` can be used, if name lookup resolves the name in the *id-expression* to either a non-static non-type member or a static member function with an explicit `this` parameter ([dcl.fct]) of some class *C*, and if either the *id-expression* is potentially evaluated or *C* is *X* or a base class of *X*, the *id-expression* is transformed into a class member access expression using `(*this)` as the *postfix-expression* to the left of the `.` operator. [*Note*: If *C* is not *X* or a base class of *X*, the class member access expression is ill-formed. — *end note*] This transformation does not apply in the template definition context ([temp.dep.type]). [*Example*: [...] - *end example*]

Strike the footnote in 7.5.4 [expr.prim.id]/2 as a drive-by-fix:

- (3.1) — as part of a class member access in which the object expression refers to the member's class ~~[Footnote: This also applies when the object expression is an implicit `(*this)` (`class.mfct.non-static`).]~~ or a class derived from that class, or

Change 7.5.5 [expr.prim.lambda]/3:

- ³ In the *decl-specifier-seq* of the *lambda-declarator*, each *decl-specifier* shall be one of `mutable`, `constexpr`, or `constexpr`. If the *lambda-declarator* contains an explicit `this` parameter ([dcl.fct]), then no *decl-specifier* in the *decl-specifier-seq* shall be `mutable`. [*Note*: The trailing requires-clause is described in [dcl.decl]. — *end note*]

Extend the example in 7.5.5.2 [expr.prim.lambda.closure]/3:

```
auto glambda = [](auto a, auto&& b) { return a < b; };
bool b = glambda(3, 3.14);                                     // OK

auto vglambda = [](auto printer) {
    return [=](auto&& ... ts) {                                // OK: ts is a function
        parameter pack
        printer(std::forward<decltype(ts)>(ts)...);

        return [=]() {
            printer(ts ...);
        };
    };
};
auto p = vglambda( [](auto v1, auto v2, auto v3)
                  { std::cout << v1 << v2 << v3; } );
auto q = p(1, 'a', 3.14);                                     // OK: outputs 1a3.14
q();                                                         // OK: outputs 1a3.14
+
+ auto fact = [](this auto self, int n) -> int {              // OK: explicit this
+     parameter
+     return (n <= 1) ? 1 : n * self(n-1);
+ };
+ std::cout << fact(5);                                       // OK: outputs 120
```

Add a new paragraph after 7.5.5.2 [expr.prim.lambda.closure]/3:

3* Given a lambda with a *lambda-capture*, the type of the explicit *this* parameter, if any, of the lambda's function call operator (possibly instantiated from a function call operator template) shall be either:

- (3*.1) — the closure type,
- (3*.2) — a class type derived from the closure type, or
- (3*.3) — a reference to a possibly cv-qualified such type.

[*Example:*

```
struct C {
    template <typename T>
    C(T);
};

void func(int i) {
    int x = [=](this auto&&) { return i; }(); // ok
    int y = [=](this C) { return i; }();      // ill-formed
    int z = [=](this C) { return 42; }();     // ok
}

— end example ]
```

Change 7.5.5.2 [expr.prim.lambda.closure]/4:

- 4 The function call operator or operator template is declared `const` ([class.mfct.non-static]) if and only if the *lambda-expression's parameter-declaration-clause* is not followed by `mutable` and the *lambda-declarator* does not contain an explicit *this* parameter.

Change 7.6.1.3 [expr.call]/1-2. The intent is to reduce the restriction on static member functions to just the ones without an explicit *this* parameter (since otherwise it would prevent the typical `x.f()` use-cases from working, since `x.f` is later defined as a prvalue). We do not need to change p2 to refer to object member functions since explicit *this* functions are also regular static member functions, and this restriction would prevent such uses.

- 1 [...] The postfix expression shall have function type or function pointer type. For a call to a non-member function or to a static member function *that does not have an explicit this parameter* ([dcl.fct]), the postfix expression shall either be an lvalue that refers to a function (in which case the function-to-pointer standard conversion ([conv.func]) is suppressed on the postfix expression), or have function pointer type.
- 2 For a call to a non-static member function, the postfix expression shall be an implicit ([class.mfct.non-static], [class.static]) or explicit class member access whose *id-expression* is a function member name, or a pointer-to-member expression selecting a function member; the call is as a member of the class object referred to by the object expression. In the case of an implicit class member access, the implied object is the one pointed to by `this`. [Note: A member function call of the form `f()` is interpreted as `(*this).f()` (see ~~[class.mfct.non-static]~~ [expr.prim.id]). — end note]

Change 7.6.1.3 [expr.call]/7:

- 7 When a function is called, each parameter ([dcl.fct]) is initialized ([dcl.init], [class.copy.ctor]) with its corresponding argument. *If the function is a static member function with an explicit this parameter and there is an implied object argument* ([over.call.func]), the list of provided arguments is preceded by the implied object argument for the purposes of this correspondence. If there is no corresponding argument, the default argument for the parameter is used. [...] If the function is a non-static member function, the `this` parameter of the function is initialized with a pointer to the object of the call, converted as if by an explicit type conversion. [*Note:* There is no access or ambiguity checking on this conversion; the access checking and disambiguation are done as part of the (possibly implicit) class member access operator. See [class.member.lookup],

[class.access.base], and [expr.ref]. — end note] When a function is called, the type of any parameter shall not be a class type that is either incomplete or abstract.

Change 7.6.1.5 [expr.ref]/6.3 - flipping the two bullets.

(6.3) If E2 is a (possibly overloaded) member function, function overload resolution ([over.match]) is used to select the function to which E2 refers. The type of E1.E2 is the type of E2 and E1.E2 refers to the function referred to by E2.

(6.3.1) — ~~Otherwise (when E2 refers to a non-static member function)~~ If E2 refers to an object member function ([dcl.fct]), E1.E2 is a prvalue. The expression can be used only as the left-hand operand of a member function call ([class.mfct]). [Note: Any redundant set of parentheses surrounding the expression is ignored ([expr.prim.paren]). — end note]

(6.3.2) — ~~If E2 refers to a static member function~~ Otherwise, E1.E2 is an lvalue.

§ 9.2.2 Wording in 9 [dcl.dcl]

In 9.3.4.6 [dcl.fct]/3, introduce *explicit-this-parameter-declaration* and *non-this-parameter-declaration* as the two kinds of *parameter-declaration*:

parameter-declaration-list:

parameter-declaration

parameter-declaration-list , *parameter-declaration*

~~*parameter-declaration*~~ *non-this-parameter-declaration*:

~~*attribute-specifier-seq*~~_{opt} *decl-specifier-seq* *declarator*

~~*attribute-specifier-seq*~~_{opt} *decl-specifier-seq* *declarator* = *initializer-clause*

~~*attribute-specifier-seq*~~_{opt} *decl-specifier-seq* *abstract-declarator*_{opt}

~~*attribute-specifier-seq*~~_{opt} *decl-specifier-seq* *abstract-declarator*_{opt} = *initializer-clause*

explicit-this-parameter-declaration:

this *non-this-parameter-declaration*

parameter-declaration:

*attribute-specifier-seq*_{opt} *explicit-this-parameter-declaration*

*attribute-specifier-seq*_{opt} *non-this-parameter-declaration*

After 9.3.4.6 [dcl.fct]/5, insert a paragraph describing where a function declaration with an explicit this parameter may appear, and renumber section.

5a An *explicit-this-parameter-declaration* shall appear only as the first *parameter-declaration* of a *parameter-declaration-list* of either:

(5a.1) — a *member-declarator* that declares a member function ([class.mem]), or

(5a.2) — a *lambda-declarator* ([expr.prim.lambda]).

5b A *member-declarator* with an *explicit-this-parameter-declaration* shall not include a *ref-qualifier* or a *cv-qualifier-seq* and shall not be declared *static* or *virtual*. [Note: Such a function is implicitly static ([class.mem]) - end note]

[Example:

```
struct C {  
    void f(this C& self);  
    template <typename Self>
```

```

    void g(this Self&& self, int);

    void h(this C) const; // error: const not allowed here
};

void test(C c) {
    c.f();           // ok: calls C::f
    c.g(42);         // ok: calls C::g<C&>
    std::move(c).g(42); // ok: calls C::g<C>
}

— end example ]

```

- 5c A function parameter declared with an *explicit-this-parameter-declaration* is an *explicit this parameter*. An explicit this parameter shall not be a function parameter pack ([temp.variadic]). An *object member function* is either a static member function with an explicit this parameter or a non-static member function.
- 5d A *non-this parameter* is a function parameter that is not the explicit this parameter. The *non-this-parameter-type-list* of a member function is the parameter-type-list of that function with the explicit this parameter, if any, omitted. [*Note*: The non-this-parameter-type-list consists of the adjusted types of all the non-this parameters. - *end note*]

Change 9.5.4 [dcl.fct.def.coroutine]/3-4:

- 3 The *promise type* of a coroutine is `std::coroutine_traits<R, P1, ..., Pn>::promise_type`, where R is the return type of the function, and P₁...P_n are the sequence of types of the *non-this* function parameters, preceded by the type of the implicit or explicit this *object* parameter ([over.match.funcs]) if the coroutine is *a non-static an object* member function. The promise type shall be a class type.
- 4 In the following, p_i is an lvalue of type P_i, where p₁ denotes **this the implicit or explicit this parameter* and p_{i+1} denotes the *ith non-this* function parameter for *a non-static an object* member function, and p_i denotes the *ith* function parameter otherwise.

§ 9.2.3 Wording in 11 [class]

Change 11.4 [class.mem]/4:

- 4 A data member or member function may be declared *static* in its *member-declaration*, in which case it is a *static member* (see [class.static]) (a *static data member* ([class.static.data]) or *static member function* ([class.static.mfct]), respectively) of the class. *A member function declared with an explicit this parameter ([dcl.fct]) is a static member function.* Any other data member or member function is a *non-static member* (a *non-static data member* or *non-static member function* ([class.mfct.non-static]), respectively). [*Note*: A non-static data member of non-reference type is a member subobject of a class object. — *end note*]

Remove 11.4.3 [class.mfct.non-static]/3 (was moved into [expr.prim.id] earlier).

Change 11.4.8.3 [class.conv.fct]/1:

- 1 *A An object* member function of a class X having no *non-this* parameters ([dcl.fct]) with a name of the form [...] specifies a conversion from X to the type specified by the *conversion-type-id*. Such functions are called *conversion functions*. A *decl-specifier* in the *decl-specifier-seq* of a conversion function (if any) shall be neither a *defining-type-specifier* nor *static*. The type of the conversion function ([dcl.fct]) is “function taking no parameter returning *conversion-type-id*”.

Change 11.4.9.2 [class.static.mfct]/2:

- 2 [*Note*: A static member function does not have a `this` pointer ([`class.this`]). — *end note*] A static member function shall not be `virtual`. There shall not be a static and a non-static member function with the same name and the same `parameter-type-list` `non-this-parameter-type-list` ([`dcl.fct`], [`over.load`]). A static member function shall not be declared `const`, `volatile`, or `const volatile`.

§ 9.2.4 Wording in 12 [\[over\]](#)

Change 12.2 [\[over.load\]/2.2](#):

- (2.2) Member function declarations with the same name, the same `parameter-type-list` `non-this-parameter-type-list` ([`dcl.fct`]), and the same trailing *requires-clause* (if any) cannot be overloaded if any of them is `a-static not an object` member function declaration (~~`{class-static}`~~) ([`dcl.fct`]). Likewise, member function template declarations with the same name, the same `parameter-type-list` `non-this-parameter-type-list`, the same trailing *requires-clause* (if any), and the same template-head cannot be overloaded if any of them is `a-static not an object` member function template declaration. The types of the implicit `or explicit object this` parameters constructed for the member functions for the purpose of overload resolution ([`over.match.funcs`]) are not considered when comparing `parameter-type-lists` `non-this-parameter-type-lists` for enforcement of this rule. In contrast, if ~~there is no static~~ every member function declaration among a set of member function declarations with the same name, the same `parameter-type-list` `non-this-parameter-type-list`, and the same *trailing requires-clause* (if any) `is an object member function`, then these member function declarations can be overloaded if they differ in the type of their implicit `or explicit object this` parameter. [*Example*: The following illustrates this distinction:

```
class X {
    static void f();
    void f();                // error
    void f() const;          // error
    void f() const volatile; // error
    void g();
    void g() const;          // OK: no static g
    void g() const volatile; // OK: no static g

+ void h(this X&, int);
+ void h(int) &&;            // OK: different this parameter type
+ void j(this const X&);
+ void j() const&;          // error: same this parameter type
+ void k(this X&);           // OK
+ void k(this X&&);          // OK
};
```

— *end example*]

Change 12.4 [\[over.match\]/1](#):

- 1 The selection criteria for the best function are the number of arguments, how well the arguments match the `parameter-type-list` of the candidate function, how well (for non-static member functions) the object matches the implicit `object this` parameter ([`dcl.fct`]), and certain other properties of the candidate function.

Change 12.4.2 [\[over.match.funcs\]/2-5](#):

- 2 So that argument and parameter lists are comparable within this heterogeneous set, a member function `that does not have an explicit this parameter` is considered to have an extra first parameter, called the *implicit object this parameter*, which represents the object for which the member function has been called. For the purposes of overload resolution, both static and non-static member functions have an implicit `or explicit object this` parameter, but constructors do not.
- 3 Similarly, when appropriate, the context can construct an argument list that contains an *implied object argument* as the first argument in the list to denote the object to be operated on.

- 4 For non-static member functions, the type of the implicit `object this` parameter is
- (4.1) — “lvalue reference to cv x” for functions declared without a *ref-qualifier* or with the *& ref-qualifier*
- (4.2) — “rvalue reference to cv x” for functions declared with the *&& ref-qualifier*

where x is the class of which the function is a member and cv is the cv-qualification on the member function declaration. [Example: For a `const` member function of class X, the extra parameter is assumed to have type “lvalue reference to `const` x”. — *end example*] For conversion functions *that are non-static member functions*, the function is considered to be a member of the class of the implied object argument for the purpose of defining the type of the implicit `object this` parameter. For non-conversion functions introduced by a *using-declaration* into a derived class, the function is considered to be a member of the derived class for the purpose of defining the type of the implicit `object this` parameter. For static member functions *that do not have an explicit this parameter*, the implicit `object this` parameter is considered to match any object (since if the function is selected, the object is discarded). [*Note*: No actual type is established for the implicit `object this` parameter of *such* a static member function, and no attempt will be made to determine a conversion sequence for that parameter ([`over.match.best`]). — *end note*]

- 5 During overload resolution, the implied object argument is indistinguishable from other arguments. The implicit `object this` parameter, however, retains its identity since no user-defined conversions can be applied to achieve a type match with it. For non-static member functions declared without a *ref-qualifier*, even if the implicit `object this` parameter is not const-qualified, an rvalue can be bound to the parameter as long as in all other respects the argument can be converted to the type of the implicit `object this` parameter. [*Note*: The fact that such an argument is an rvalue does not affect the ranking of implicit conversion sequences. — *end note*]

Change 12.4.2.2.2 [`over.call.func`]/3 and adjust the corresponding footnote, and add an example:

- 3 Because of the rules for name lookup, the set of candidate functions consists (1) entirely of non-member functions or (2) entirely of member functions of some class T. In case (1), the argument list is the same as the expression-list in the call. In case (2), the argument list is the *expression-list* in the call augmented by the addition of an implied object argument as in a qualified function call. If the keyword `this` is in scope and refers to class τ , or a derived class of τ , then the implied object argument is `(*this)`. If the keyword `this` is not in scope or refers to another class, then a contrived object of type τ becomes the implied object argument.
- ¹²³ If the argument list is augmented by a contrived object and overload resolution selects one of the *non-static object* member functions of τ , the call is ill-formed.

[*Example*:

```
struct C {
    void a();
    void b() {
        a(); // ok, (*this).a()
    }

    void f(this const C&);
    void g() const {
        f(); // ok: (*this).f()
        f(*this); // error: no viable candidate for (*this).f(*this)
        this->f(); // ok
    }

    static void h() {
        f(); // error: contrived object argument, but overload resolution picked an
            // object member function
        f(C{}); // error: no viable candidate
        C{}.f(); // ok
    }
};
```

— *end example*]

- 119 An implied object argument must be contrived to correspond to the implicit **object this** parameter attributed to member functions during overload resolution. It is not used in the call to the selected function. Since the member functions all have the same implicit object parameter, the contrived object will not be the cause to select or reject a function.

Add to 12.4.2.2.3 [\[over.call.object\]/3](#):

- 3 The argument list submitted to overload resolution consists of the argument expressions present in the function call syntax preceded by the implied object argument (**E**). [*Note*: When comparing the call against the function call operators, the implied object argument is compared against **either** the implicit **or explicit object this** parameter of the function call operator. When comparing the call against a surrogate call function, the implied object argument is compared against the first parameter of the surrogate call function. The conversion function from which the surrogate call function was derived will be used in the conversion sequence for that parameter since it converts the implied object argument to the appropriate function pointer or reference required by that first parameter. — *end note*]

Change the note in 12.4.2.3 [\[over.match.oper\]/3.4](#):

- (3.4.5) [*Note*: A candidate synthesized from a member candidate has its implicit **or explicit object this** parameter as the second parameter, thus implicit conversions are considered for the first, but not for the second, parameter. — *end note*]

Change the note in 12.4.2.5 [\[over.match.copy\]/2](#):

- 2 In both cases, the argument list has one argument, which is the initializer expression. [*Note*: This argument will be compared against the first parameter of the constructors and against the implicit **or explicit object this** parameter of the conversion functions. — *end note*]

Change the note in 12.4.2.6 [\[over.match.conv\]/2](#):

- 2 The argument list has one argument, which is the initializer expression. [*Note*: This argument will be compared against the implicit **or explicit object this** parameter of the conversion functions. — *end note*]

Change the note in 12.4.2.7 [\[over.match.ref\]/2](#):

- 2 The argument list has one argument, which is the initializer expression. [*Note*: This argument will be compared against the implicit **or explicit object this** parameter of the conversion functions. — *end note*]

Change 12.4.4.2 [\[over.best.ics\]/4](#):

- 4 However, if the target is
- (4.1) — the first parameter of a constructor or
- (4.2) — the implicit **or explicit object this** parameter of a user-defined conversion function
- and the constructor or user-defined conversion function is a candidate by [...]

Change 12.4.4.2 [\[over.best.ics\]/7](#):

- 7 In all contexts, when converting to the implicit **object this** parameter or when converting to the left operand of an assignment operation only standard conversion sequences are allowed. [*Note*: When converting to the **explicit this** parameter, if any, user-defined conversion sequences are allowed. — *end note*]

Change 12.4.4.2.3 [\[over.ics.user\]/1](#):

- 1 If the user-defined conversion is specified by a conversion function, the initial standard conversion sequence converts the source type to the implicit `or explicit object this` parameter of the conversion function.

Change 12.4.4.2.5 [\[over.ics.ref\]/3](#):

- 3 Except for an implicit `object this` parameter, for which see [\[over.match.funcs\]](#), an implicit conversion sequence cannot be formed if it requires binding an lvalue reference other than a reference to a non-volatile const type to an rvalue or binding an rvalue reference to an lvalue other than a function lvalue. [*Note*: This means, for example, that a candidate function cannot be a viable function if it has a non-const lvalue reference parameter (other than the implicit `object this` parameter) and the corresponding argument would require a temporary to be created to initialize the lvalue reference (see [\[dcl.init.ref\]](#)). — *end note*]

Change 12.4.4.3 [\[over.ics.rank\]/3.2.3](#):

- (3.2) Standard conversion sequence S1 is a better conversion sequence than standard conversion sequence S2 if
 - (3.2.3) — S1 and S2 include reference bindings ([\[dcl.init.ref\]](#)) and neither refers to an implicit `object this` parameter of a non-static member function declared without a *ref-qualifier*, and S1 binds an rvalue reference to an rvalue and S2 binds an lvalue reference

Change 12.6 [\[over.oper\]/7](#):

- 7 An operator function shall either be `a-non-static an object` member function or be a non-member function that has at least one parameter whose type is a class, a reference to a class, an enumeration, or a reference to an enumeration.

Change 12.6.2 [\[over.unary\]/1](#):

- 1 A *prefix unary operator function* is a function named `operator@` for a prefix *unary-operator @* ([\[expr.unary.op\]](#)) that is either `a-non-static an object` member function ([\[class.mfct\]](#)) with no `non-this` parameters or a non-member function with one parameter.

Change 12.6.3 [\[over.binary\]/1](#):

- 1 A *binary operator function* is a function named `operator@` for a binary operator `@` that is either `a-non-static an object` member function ([\[class.mfct\]](#)) with one `non-this` parameter or a non-member function with two parameters.

Change 12.6.3.2 [\[over.ass\]/1](#):

A *simple assignment operator function* is a binary operator function named `operator=`. A simple assignment operator function shall be `a-non-static an object` member function.

Change 12.6.4 [\[over.call\]/1](#):

- 1 A *function call operator function* is a function named `operator()` that is `a-non-static an object` member function with an arbitrary number of parameters.

Change 12.6.5 [\[over.sub\]/1](#):

- 1 A *subscripting operator function* is a function named `operator[]` that is `a-non-static an object` member function with exactly one `non-this` parameter.

Change 12.6.6 [\[over.ref\]/1](#):

- 1 A *class member access operator function* is a function named `operator->` that is `a-non-static an object` member function taking no `non-this` parameters.

Change 12.6.7 [\[over.inc\]](#)/1:

- 1 An *increment operator function* is a function named `operator++`. If this function is ~~a non-static~~ **an object** member function with no **non-this** parameters, or a non-member function with one parameter, it defines the prefix increment operator `++` for objects of that type. If the function is ~~a non-static~~ **an object** member function with one **non-this** parameter (which shall be of type `int`) or a non-member function with two parameters (the second of which shall be of type `int`), it defines the postfix increment operator `++` for objects of that type.

§ 9.3 Feature-test macro `[tab:cpp.predefined.ft]`

Add to 15.11 [\[cpp.predefined\]](#)/table 17 (`[tab:cpp.predefined.ft]`):

`__cpp_explicit_this_parameter` with the appropriate value.

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§ 11 References

[EffCpp] Scott Meyers. Effective C++, Third Edition.

<https://www.aristeia.com/books.html>

[P0798R0] Simon Brand. 2017. Monadic operations for `std::optional`.

<https://wg21.link/p0798r0>

[P0798R3] Simon Brand. 2019. Monadic operations for `std::optional`.

<https://wg21.link/p0798r3>

[P0826R0] Agustín Bergé. 2017. SFINAE-friendly `std::bind`.

<https://wg21.link/p0826r0>

[P0839R0] Richard Smith. 2017. Recursive Lambdas.

<https://wg21.link/p0839r0>

[P0847R0] Gašper Ažman, Simon Brand, Ben Deane, Barry Revzin. 2018. Deducing this.

<https://wg21.link/p0847r0>

[P0847R1] Gašper Ažman, Simon Brand, Ben Deane, Barry Revzin. 2018. Deducing this.

<https://wg21.link/p0847r1>

[P0847R2] Gašper Ažman, Simon Brand, Ben Deane, Barry Revzin. 2019. Deducing this.

<https://wg21.link/p0847r2>

[P0929R2] Jens Maurer. 2018. Checking for abstract class types.

<https://wg21.link/p0929r2>

[P1169R0] Barry Revzin, Casey Carter. 2018. static operator().

<https://wg21.link/p1169r0>

[P1221R1] Jason Rice. 2018. Parametric Expressions.

<https://wg21.link/p1221r1>